

AYUSH PRAKASH TIWARI

Full Stack Developer | Backend & Systems Engineer

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Summary

Software Engineer focused on **backend systems** and **AI-powered applications**, with experience building **Spring Boot** and **FastAPI** services, **RAG pipelines**, **REST APIs**, and containerized applications using **Docker**.

Experience

Backend Developer Intern | Assembly

April 2026 – Aug 2026

- Implemented **backend modules and APIs** for script breakdown and production report automation, handling **100+ scene breakdowns** with real-time dashboard updates.
- Designed **smart scheduling** to optimize cast and location usage, improving operational efficiency by **40%**.

Open Source Contributor | Eclipse Foundation

[Link](#) 2026

- Built a **Quick Export** feature for Thing Description affordances in eclipse-editord.
- Ran **functional & regression testing** on IoT workflows, ensuring zero defects and **impacting 1,000+ developers and professionals**.
- Partnered with maintainers via **code reviews** and **CI pipelines** on coding standards.

Technical Skills

Backend: Java, Spring Boot, Python, FastAPI, REST APIs, Microservices, JWT Auth, Event-Driven Design

Databases & Cloud: MongoDB, MySQL, Redis, Docker, CI/CD

Frontend: React.js, TypeScript, JavaScript, Responsive Dashboard UI

Systems & AI: System Design, Real-Time Pipelines, RAG, LangChain, Deep Learning

Projects

Autonomous Code Review Agent | *LangGraph, Java 21, Spring Boot, Python, FastAPI, Docker*

[Link](#)

- Designed an **event-driven, multi-agent backend** (6 agents) to autonomously review GitHub PRs.
- Auto-detected **security, performance, and architecture** issues like SQLi, race conditions, N+1 queries.
- Verified fixes in an **isolated Docker sandbox** and posted results via **GitHub API** integration.

Self-Healing RAG Pipeline | *Python, FastAPI, Spring Boot, TypeScript, Qdrant, MongoDB, Docker*

[Link](#)

- Architected a **3-stage low-latency retrieval pipeline** with query rewriting and grounding validation.
- Built a **FastAPI microservice** and **Spring Boot API gateway** with OAuth2/JWT auth.
- Shipped a **Dockerized, cloud-deployable monorepo** with pluggable LLM/vector-store abstraction.

Image Encryption Vault | *React 18.3, TypeScript 5.5, Vite, Tailwind CSS*

[Link](#)

- Built a **client-side encryption app** with XOR-based pixel encryption and zero server uploads.
- Deployed on **Vercel**, 95.6% TypeScript, with a responsive dashboard-style UI.

Education

Jaypee University of Engineering and Technology

B.Tech in Computer Science and Engineering

Aug 2023 – 2027

Guna, Madhya Pradesh

Achievements

- Secretary**, Computer Society of India – led technical development for student chapter.
- Runner-up**, Walmart Sparkathon national hackathon (500+ teams).
- Built and Organized **2 platforms** (500+ users) for 48-hr hackathon.